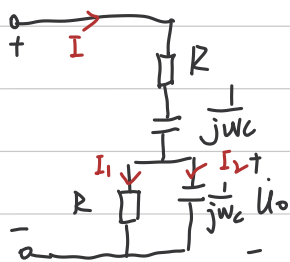


2.3.12

当 $f_0 = \frac{1}{2\pi RC}$ $\omega = 2\pi f_0 = \frac{1}{RC}$ $\therefore \frac{1}{j\omega C} = j \frac{1}{\frac{1}{RC}C} = \frac{R}{j} = -jR$

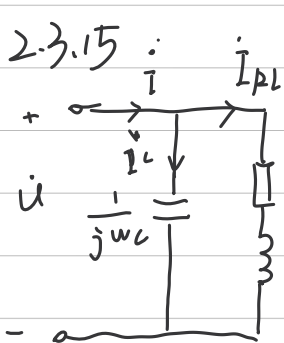


$$\begin{cases} I = I_1 + I_2 \\ u_o = I_1 R = I_2 \frac{1}{j\omega C} \Rightarrow I_1 j = I_2 \\ u_i = j \left(R + \frac{1}{j\omega C} + \frac{R}{j\omega C R + 1} \right) \end{cases} \quad \therefore \frac{u_o}{u_i} = \frac{I_1 R}{j \left(R + \frac{1}{j\omega C} + \frac{R}{j\omega C R + 1} \right)}$$

$$= \frac{2I_1}{3I(1-j)} = \frac{2}{3(1-j)} = \frac{1}{3} \angle 0^\circ$$

证毕

2.3.15



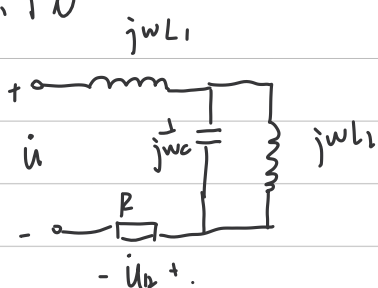
$$\begin{cases} I = I_c + I_L \\ I_c \frac{1}{j\omega C} = u \\ I_L (R + j\omega L) = u \end{cases}$$

总阻抗 $Z_{总} = \frac{\frac{1}{j\omega C} (R + j\omega L)}{\frac{1}{j\omega C} + R + j\omega L} = \frac{R + j\omega L}{1 + Rj\omega C - \omega^2 LC} = \frac{(R + j\omega L)(1 - \omega^2 LC - Rj\omega C)}{(1 - \omega^2 LC)^2 + R^2 \omega^2 C^2}$

虚部为0.得. $(1 - \omega^2 LC) \omega L - R^2 \omega C = 0 \quad \therefore f = \frac{\omega}{2\pi} \approx 62 \text{ Hz}$

由此三式得 $\begin{cases} I = 4.46 \text{ A} \\ I_c = 4.97 \angle 90^\circ \text{ A} \\ I_L = 6.67 \angle -48.12^\circ \text{ A} \end{cases}$

2.3.16



当 $\omega = 1000 \text{ rad/s}$ 时. $\therefore u_o = 0 \quad \therefore I = 0$

发生并联谐振. $L_2 \omega^2 C = 1 \quad \therefore L_2 = \frac{1}{1000^2 \times 10^{-6}} \text{ H} = 1 \text{ H}$

当 $\omega = 2000 \text{ rad/s}$ 时. $u_o = u = 10 \text{ V} \quad \therefore$ 发生串联谐振.

$$j\omega L_1 + \frac{j\omega L_2 \cdot \frac{1}{j\omega C}}{\frac{1}{j\omega C} + j\omega L_2} = 0$$

$$\therefore L_1 = \frac{L_2}{\omega^2 L_2 C - 1} = 0.33 \text{ H}$$

